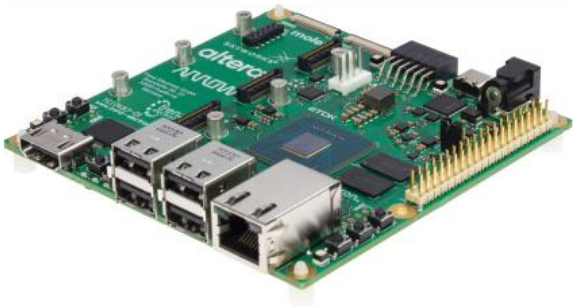


Dual-Camera Picture-in-Picture Reference design



The Dual-Camera Picture-in-Picture (PiP) reference design demonstrates the Altera™ Agilex™-5 SoC FPGA's MIPI and video-processing capabilities on the Arrow AXE5-Falcon Development Kit.



The first camera streams 1920x1080 RAW10 images at 30fps. The second camera stream images and scales them down to 640x480 RAW10 at 30fps, processed through a Video ISP and triple-buffered in DDR4. The first stream fills the HD background; the other becomes a Picture-in-Picture window bounced around the display by a Nios® V/m microcontroller, with HDMI output at 60fps.

Built with Quartus Prime Pro 26.1, the design ships ready-to-flash with full FPGA and Nios V sources on Arrow's GitHub Wiki.

Partners:

altera An Intel Company · micron · samtec · SKYWORKS · TDK · trenz electronic

Key Features: Arrow AXE5-Falcon Kit

- A5EE013BB23BE6SCS
  - * 138 kLE
  - * Dual A55 SoC
  - * 376 multipliers 18x19
  - * 4.93 Peak INT8 (TOPS)
  - * 6.99 Mb (M20K) internal memory
  - * MIPI D-PHY interface
  - * LVDS pairs at 1.250 Mbps
  - * Secure Device Manager
- DDR4: 2x 1Gbit x16 (1Gb x 32)
- Nios® V/m soft microcontroller + on-chip RAM
- 2x MIPI CSI 22-pin connectors (dual camera input)
- HDMI Output Interface
- USB 2.0
- 1x 1Gb Ethernet
- MicroSD Card
- 4x RGB LEDs, Push Buttons, DIP switches
- On-board USB Blaster III Programmer
- 5V Power via USB-C or Barrel Jack
- Size: 10cm x 10cm

Ordering number: AXE5-Falcon
Technical details: GitHub page